Tyrone Marhguy

Computer Engineering | Architecture, RTL, FPGA & ASIC
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Education

University of Pennsylvania Philadelphia, PA
B.S.E. in Computer Engineering Expected May 2028
Coursework: Computer Organization & Design, VLSI, Digital Logic, Embedded Systems

Experience

Hardware & Firmware Engineer May – Aug. 2026
Vero Electric Philadelphia, PA

- Designed and routed a **BQ79616 16-channel battery-monitoring front end** for a **240 kWh / 120 kW** energy-storage system; closed PCB ERC/DRC with zero errors.
- Developed embedded C/C++ fault detection and SPI/I2C acquisition firmware; supported board bring-up with oscilloscopes and logic analyzers.

Open-Source EDA Contributor 2026 – Present
OpenROAD, Verilator, LibreLane, OpenFPGA C++ / Python

- Fixed OpenROAD's LEF58 parser and Verilator's Linux peak-memory accounting; shipped Yosys compatibility and synthesis-reporting fixes in **LibreLane 3.0.8 and 3.0.10**.

Software Engineer Intern May – Aug. 2026
Aragorn AI Remote

- Contributed to production backend/API integrations for AI services; debugged cross-service failures and documented service interfaces.

Selected Hardware Projects

Tomato | Custom 32-bit CPU SystemVerilog / FPGA / KiCad Demo | Code

- Designed a three-source datapath with **dual programmable LUT3 planes**, 8-way carry/flag selection and **524,288 ALU control configurations**; implemented microcoded control and an assembler.
- Built a complete FPGA computer that boots assembly-written Tomato OS over HDMI; fabricated and soldered an 8-bit 74xx ALU slice, with formal proofs at 1/8/32-bit widths.
- Recorded a **130-billion-vector Verilator ALU regression** against a reference model; checked in the verification harness and reproduction commands.

RISC-V CPU | riscv64xO3 SystemVerilog / AXI4-Lite Docs | Code

- Implemented an **RV64IM scalar pipeline** and AXI4-Lite instruction/data masters; developing 2-wide out-of-order execution. Added Verilator/cocotb ISA tests and Yosys/OpenLane2 feasibility flows.

Pineapple GPU | Programmable 3D Graphics Verilog / Artix-7 Code

- Built programmable vertex/fragment shaders, rasterization, depth and texture pipelines, and **320×180 double buffering**; rendered interactive scenes on a Nexys A7 using Yosys/nextpnr.

FPGA Networking & NASDAQ ITCH SystemVerilog / cocotb UDP | ITCH

- Built a **100 Mbps RMII/MAC/IPv4/UDP** pipeline and ITCH 5.0 parser with a 512-entry BRAM order book; closed timing at **100 MHz** and checked 10M messages against a Python reference with zero BBO mismatches.

ASIC & Full-Custom Datapaths Sky130 / SPICE MAC | SRAM

- Designed a two-stage **BF16-to-FP32 MAC** with cocotb verification and LibreLane flow configuration; built a transistor-level **16×4 6T SRAM** with StrongARM sensing and NGSpice readback sweeps.

Technical Skills

RTL / Architecture: SystemVerilog, Verilog, VHDL; CPU/GPU datapaths, pipelining, memory systems, CDC
Verification: UVM, SVA, cocotb, formal verification, Verilator, Python, C/C++
Implementation: Vivado, Yosys, OpenROAD, LibreLane/OpenLane, STA, P&R, DRC/LVS, SPICE, KiCad
Interfaces / Tools: Ethernet, RMII, AXI4-Lite, UART, SPI, I2C; Linux, Tcl, Bash, Git